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1 Known Results

$$Y_{it} = a_i + X_i' F_t + \varepsilon_{it} \tag{1}$$

$$EY_i = m(X_i) \quad \text{or} \quad EY_i = X_i'\gamma \tag{2}$$

So

$$Y_{it} = m(X_i) + X_i'(F_t - EF) + \varepsilon_{it} \tag{3}$$

$$a_i = m(X_i) - X_i' EF = X_i'(\gamma - EF) * \quad \text{under} \quad H_0 \tag{4}$$

Assumption 1. *We make the following assumptions, for $t = 1, \ldots, T, \ldots$ and $\iota \in \mathcal{I}$,*

1. $Y_t(\iota) = a(\iota) + X(\iota)' F_t + \varepsilon_t(\iota)$.
2. *F_t are iid over t. For all $\iota \in \mathcal{I}$, we have $\varepsilon_t(\iota)$ is a m.d.s.*
3. *Random sampling: for $i = 1, \ldots, n$, we observe $(Y_{it}, X_{it}) = (Y_t(\iota_i), X_t(\iota_i))$. This follows from Andrews, 2005 and Gagliardini, Ossola, and Scaillet, 2016.*
4. *$X(\iota) \in \mathcal{X}$ a compact subset of $\mathbb{R}^k$ for all ι. For all ι, $X(\iota)$ is a random vector with the same probability density $f(x)$ which is bounded away from 0 on $\mathcal{X}$.*

We can estimate with both cross-sectional and time-series regression under these settings.

We follow the following proof method:

1. Uniform convergence result $\sup_x |x'(\hat{\gamma} - \gamma)| = o_p(\frac{1}{\sqrt{nT}})$.
2. $\sup_x |\hat{m}(x) - m(x)| = o_p(\frac{1}{\sqrt{nT}})$.
3. Asymptotic linearization of $\hat{m}$ and $x'\hat{\gamma}$.
4. Construct test based on

2 About Parametric Estimation Error

Let F_D, Y_D be the demeaned version of F, Y. Then

$$\hat{X}_i = \left(F_D' F_D\right)^{-1} F_D' Y_{D,i} \quad \hat{X} = Y_D F_D \left(F_D' F_D\right)^{-1} \tag{5}$$

Lemma 2.1 ($\hat{\gamma}$ under H_0)**.**

$$\begin{aligned}
\hat{\gamma} &= \left(\hat{X}\hat{X}\right)^{-1} \hat{X}'\bar{Y}_i && (6)\\
&= \left(\frac{1}{n}\hat{X}\hat{X}\right)^{-1} \frac{1}{n}\left[\hat{X}'\bar{Y}\right] && (7)\\
&= \gamma + \bar{F} - EF + \left(\frac{1}{n}\hat{X}\hat{X}\right)^{-1} \frac{1}{n}\hat{X}\left(\bar{\varepsilon} - \left(\hat{X} - X\right)\left[\gamma + \bar{F} - EF\right]\right) && (8)
\end{aligned}$$

We want to calculate under the $H_0 : E(Y_i \mid X_i = x) = x'\gamma$, the residual:

$$\begin{aligned}
\hat{e}_i &= \bar{Y}_i - \hat{X}_i\hat{\gamma} && (9)\\
&= X_i'[\gamma + \bar{F} - EF] + \bar{\varepsilon}_i - X_i'\hat{\gamma} + \left(\hat{X}_i - X_i\right)'\gamma && (10)\\
&= X_i'[\gamma + \bar{F} - EF] + \bar{\varepsilon}_i - X_i'\left(\hat{X}_i\hat{X}_i\right)^{-1}\hat{X}_i'\bar{Y}_i + \left[\left(F_D' F_D\right)^{-1} F_D'\bar{\varepsilon}_i\right]'\gamma && (11)\\
&= \bar{\varepsilon}_i + X_i'\left[\hat{\gamma} - \gamma - \bar{F} + EF\right] + \left[\left(F_D' F_D\right)^{-1} F_D'\bar{\varepsilon}_i\right]'\gamma && (12)\\
& && (13)
\end{aligned}$$

We can show that $\sup_x x'(\hat{\gamma} - \gamma) = O_p(1/\sqrt{nT})$.

3 Asymptotics About the Local Linear Estimator

3.1 Local Linear Estimator and Notations

Under H_0 and Assumptions hold, let $m_0(x, \gamma) = x'\gamma$, because

$$E(Y_i) = X_i' E(F) \quad \bar{Y}_i = X_i' \bar{F} + \bar{\varepsilon}_i \tag{14}$$

$$E(Y_i) = m_0(X_i, \gamma) \quad \bar{Y}_i = \bar{m}_0(X_i, \gamma) + \bar{\varepsilon}_i \tag{15}$$

Across t, we can assume that $\varepsilon_{i,t}$ are either, independent, stationary and β-mixing(See Carrasco and Chen on GARCH), m.d.s, etc.

We have

$$\bar{m}_0(X_i, \gamma) = m_0(X_i, \gamma) + X_i'\big(\bar{F} - E(F)\big)$$

Let $\bar{M} = \begin{bmatrix} \bar{m}_0(X_1, \gamma) \\ \vdots \\ \bar{m}_0(X_n, \gamma) \end{bmatrix}$ and $\bar{\varepsilon} = \begin{bmatrix} \bar{\varepsilon}_1 \\ \vdots \\ \bar{\varepsilon}_n \end{bmatrix}$.

For any variable a_{it}, $\bar{a}_i = \frac{1}{T}\sum_t a_{it}$. In the first step we use OLS,

$$\hat{X}_i = (F'F)^{-1} F' Y_i \quad \text{so} \quad \hat{X} = Y_{(n\times T)} F \left(F'F\right)^{-1}$$

For $x \in \operatorname{supp}(X) \subset \mathbb{R}^k$, let local linear estimator be $\hat{m}(x) = e_1^\top \left(\hat{X}_x' \hat{W}_x \hat{X}_x\right)^{-1} \hat{X}_x' \hat{W}_x \bar{Y}$ where $e_1^\top = (1, 0, \ldots, 0)'$

$$\hat{X}_x = \left[\mathbf{1} \quad \left(\hat{X} - \mathbf{1} \otimes x'\right)\right] \quad \hat{W}_x = \operatorname{diag}\left(K_h\left(\hat{X}_1 - x\right), \ldots, K_h\left(\hat{X}_n - x\right)\right)$$

Let $\theta(x) = (m(x), Dm(x))'$.

Assumption 2. *We restrict the alternative by assuming $m(x) \in \mathcal{C}^2$. Let $K(\cdot)$ be a symmetric probability density function with compact support. $K_h(\cdot) = \frac{1}{h} K\left(\frac{\cdot}{h}\right)$.*

Then we have

$$\begin{aligned}\hat{\theta}(x) - \theta(x) &= \left(\hat{X}_x' \hat{W}_x \hat{X}_x\right)^{-1} \hat{X}_x' \hat{W}_x \bar{Y} \\ &= \hat{S}_x^{-1} \hat{X}_x' \hat{W}_x \left(\bar{M} + \bar{\varepsilon} - \hat{X}_x \theta(x)\right) \\ &= \hat{S}_x^{-1} \hat{X}_x' \hat{W}_x \left(M - X_x' \theta(x) + X\left(\bar{F} - EF\right) + \left(\hat{X}_x - X_x\right)' \theta(x) + \bar{\varepsilon}\right)\end{aligned}$$

where $\hat{S}_x = \hat{X}_x' \hat{W}_x \hat{X}_x$.

3.2 Pointwise Asymptotics

Lemma 3.1. *We have the following decompositions:*

1. $\hat{X}_x = X_x + R_1$ *where* $R_1 = \begin{bmatrix} \mathbf{0} & \hat{X} - X \end{bmatrix} = O_p\left(\frac{1}{\sqrt{T}}\right)$;

2. $\hat{W}_x = W_x + R_2$, *where*

$$R_2 = \operatorname{diag}\left(\frac{1}{h^2} K'\left(\frac{X_1 - x}{h}\right)\left(\hat{X}_1 - X_1\right), \ldots, \frac{1}{h^2} K'\left(\frac{X_n - x}{h}\right)\left(\hat{X}_n - X_n\right)\right) \tag{16}$$

$$= O_p\left(\frac{1}{h^2\sqrt{T}}\right) \tag{17}$$

3. $\frac{1}{n}\hat{S}_x = S_x + Remainders.$

Proof. $\hat{X}_x = \begin{bmatrix} \mathbf{1}_n & \hat{X} - \mathbf{1}_n x' \end{bmatrix} = \begin{bmatrix} \mathbf{1}_n & X - \mathbf{1}_n x' \end{bmatrix} + \begin{bmatrix} 0 & \hat{X} - X \end{bmatrix} = X_x + O_p\left(\frac{1}{\sqrt{T}}\right)$.

Assuming K_h has bounded support and bounded first-order derivative, then by Taylor expansion we have the estimate for $\hat{W}_x$. □

Lemma 3.2. *For* $k = 1$ *we will have*

$$\hat{X}_x' \hat{W}_x \hat{X}_x = \begin{bmatrix} \sum K_h\left(\hat{X}_j - x\right) & \sum K_h\left(\hat{X}_j - x\right)\left(\hat{X}_j - x\right) \\ * & \sum K_h\left(\hat{X}_j - x\right)\left(\hat{X}_j - x\right)^2 \end{bmatrix} \tag{18}$$

Normalize by $1/n$ *we have*

$$\frac{1}{n}\hat{S}_x = \begin{bmatrix} f(x) + O_p(h^2 + \frac{1}{\sqrt{nh}} + \frac{1}{h^2\sqrt{T}}) & * \\ * & \dagger \end{bmatrix} \tag{19}$$

The $*$ *part is*

$$* \quad = o_p(1) + O_p(\frac{1}{\sqrt{T}}) + O_p(\frac{1}{h\sqrt{T}}) + O_p\left(\frac{1}{h^2 T}\right) \tag{20}$$

The $\dagger$ *part is*

$$\dagger = h^2\kappa_2 f(x) + remainder \tag{21}$$

3.3 Uniform Convergence Rate of Local Linear Estimator

Hansen 2008. Jenish 2012. Li, Lu and Linton. Here the estimation error is

$$U_n(x) = X\left(\bar{F} - EF\right) + \left(\hat{X}_x - X_x\right)\theta(x) + \bar{\varepsilon} \tag{22}$$

$$\text{var}(U) = \frac{1}{T}X_i'\Sigma_F X_i + \frac{1}{T}\Sigma_\varepsilon + (F'F)^{-1}\Sigma_\varepsilon(F'F)^{-1} \tag{23}$$

we define the local linear estimator as

$$\hat{\theta}(x) - E\hat{\theta}(x) + E\hat{\theta}(x) - \theta(x) = \left(\hat{S}_x\right)^{-1} U_x \tag{24}$$

Remark. For the cross-sectional correlation in estimation, see Gagliardini, Ossola, and Scaillet, 2016 P.1030 Lemma 3.

Lemma 3.3. *Under the assumptions, we can bound the items:*

1. $\sup_x \hat{S}_x = O_p(a_{1,n})$*; Lemma 2.9 in Chapter 36 might come in handy.*
2. $\sup_x \left\|\hat{S}_x^{-1}\right\| = O_p(1)$

Proof. By showing that $1/\inf_x \lambda_{min}(\hat{S}_x) \leq B$ for some B has probability approaching 1. See Jenish 2012 Lemma D.3. □

Lemma 3.4. $\sup_x \left\|\hat{U}_x - E\hat{U}_x\right\| = O_p(a_{2,n})$ *and* $\sup_x \left\|E\hat{U}_x\right\| = O_p(a_{3,n})$.

Theorem 3.5. *We have the uniform convergence rate of the local linear estimator:*

$$\sup_{x\in\mathcal{X}} |\hat{m}(x) - E\hat{m}(x)| = O_p(c_{n,T}) \tag{25}$$

where $c_{n,T}$ =???

Proof.

$$\sup_{x\in\mathcal{X}} |\hat{m}(x) - E\hat{m}(x)| \leq \sup_{x} \left\| \hat{S}_x^{-1} \right\| \big(O_p(a_{2,n}) + O_p(a_{3,n}) \big) \tag{26}$$

□

Hansen 2008's Proof.

Proof. Step 1, we do the truncation as in Hansen 2008. Then we separate the domain of $\mathcal{X}$ into balls $\mathcal{X}_i = \{x : \|x - x_i\| \leq c_n\}$, by compactness, there are finitely many such balls $N \leq c_n'$ covering $\mathcal{X}$, let them be $\big(\mathcal{X}_j\big)$.

$$\sup_{x\in\mathcal{X}} (\hat{m}(x) - E\hat{m}(x)) \leq N_{c_n'} \max_j \sup_{x\in\mathcal{X}_j} (\hat{m}(x) - E\hat{m}(x)) \tag{27}$$

□

3.4 Specification Test

We show that this fits into the framework of Escanciano. We calculate the following quantities:

$$\hat{\varepsilon}_{j,0} = \frac{Y_j - m_0\big(X_j, \hat{\gamma}\big)}{\hat{\sigma}(X_j)} \quad \text{and} \quad \hat{\varepsilon}_j = \frac{Y_j - \hat{m}\big(X_j\big)}{\hat{\sigma}(X_j)}$$